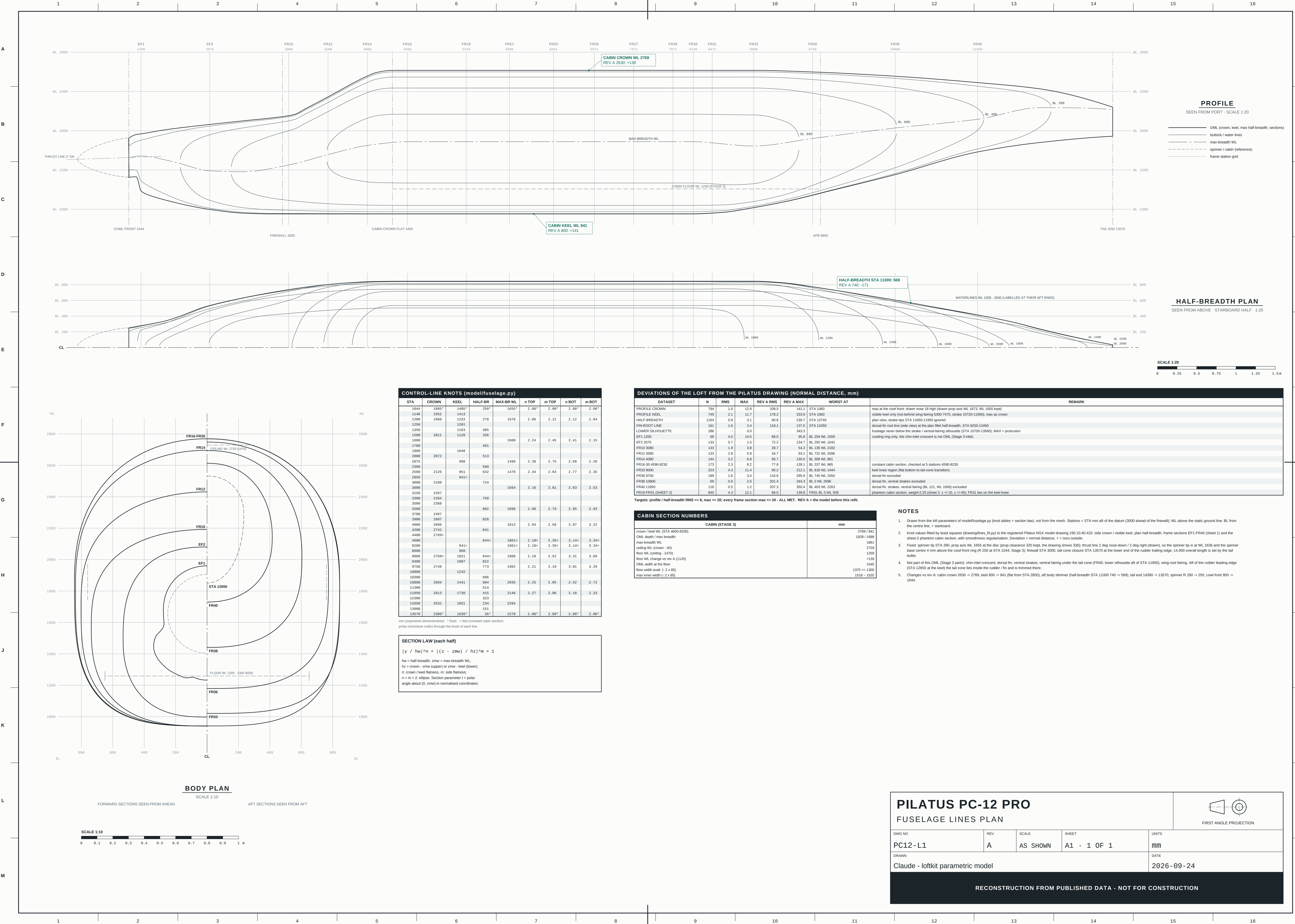




| CONTROL-LINE KNOTS (model/fuselage.py) |       |       |         |           |       |       |       |       |  |
|----------------------------------------|-------|-------|---------|-----------|-------|-------|-------|-------|--|
| STA                                    | CROWN | KEEL  | HALF-BR | MAX-BR WL | n TOP | m TOP | n BOT | m BOT |  |
| 1844                                   | 1865* | 1405* | 258*    | 1655*     | 2.06* | 2.06* | 2.06* | 2.06* |  |
| 1148                                   | 1852  | 1413  |         |           |       |       |       |       |  |
| 1209                                   | 1906  | 1223  | 278     | 1678      | 2.06  | 2.12  | 2.12  | 2.04  |  |
| 1259                                   |       | 1261  |         |           |       |       |       |       |  |
| 1359                                   |       | 1163  | 305     |           |       |       |       |       |  |
| 1509                                   | 2911  | 1120  | 336     |           |       |       |       |       |  |
| 1609                                   |       |       |         |           |       |       |       |       |  |
| 1769                                   |       |       | 461     | 1608      | 2.24  | 2.45  | 2.41  | 2.15  |  |
| 1809                                   |       | 1848  |         |           |       |       |       |       |  |
| 2099                                   | 2972  |       | 513     |           |       |       |       |       |  |
| 2399                                   |       | 998   |         | 1490      | 2.38  | 2.75  | 2.68  | 2.28  |  |
| 2509                                   | 2126  | 951   | 589     |           |       |       |       |       |  |
| 2659                                   |       | 941=  | 632     | 1478      | 2.34  | 2.83  | 2.77  | 2.35  |  |
| 3099                                   | 2180  |       | 724     |           |       |       |       |       |  |
| 3899                                   |       |       |         | 1564      | 2.16  | 2.81  | 2.83  | 2.53  |  |
| 3159                                   | 2207  |       |         |           |       |       |       |       |  |
| 3399                                   | 2284  |       | 769     |           |       |       |       |       |  |
| 3599                                   | 2385  |       |         |           |       |       |       |       |  |
| 3589                                   |       |       | 802     | 1696      | 2.08  | 2.79  | 2.95  | 2.93  |  |
| 3709                                   | 2497  |       |         |           |       |       |       |       |  |
| 3999                                   | 2697  |       | 828     |           |       |       |       |       |  |
| 4999                                   | 2695  |       |         | 1813      | 2.04  | 2.58  | 3.07  | 3.22  |  |
| 4209                                   | 2742  |       | 841     |           |       |       |       |       |  |
| 4409                                   | 2769= |       |         |           |       |       |       |       |  |
| 4599                                   |       | 941=  | 844=    | 1961=     | 2.18= | 2.35= | 3.14= | 3.34= |  |
| 6209                                   |       |       |         | 1861=     | 2.18= | 2.35= | 3.14= | 3.34= |  |
| 8609                                   |       | 960   |         |           |       |       |       |       |  |
| 9099                                   | 2769= | 1821  | 844=    | 1806      | 2.18  | 2.52  | 3.31  | 3.69  |  |
| 9499                                   |       | 1997  | 822     |           |       |       |       |       |  |
| 9759                                   | 2748  |       | 773     | 1902      | 2.21  | 2.19  | 3.01  | 3.29  |  |
| 10909                                  |       | 1242  |         |           |       |       |       |       |  |
| 10309                                  |       |       | 686     |           |       |       |       |       |  |
| 10609                                  | 2694  | 1441  | 604     | 2035      | 2.25  | 2.05  | 2.52  | 2.72  |  |
| 11309                                  |       |       | 514     |           |       |       |       |       |  |
| 11859                                  | 2613  | 1730  | 415     | 2146      | 2.27  | 2.06  | 2.18  | 2.33  |  |
| 12309                                  |       |       | 323     | 2294      |       |       |       |       |  |
| 12659                                  | 2532  | 1851  | 234     |           |       |       |       |       |  |
| 13999                                  |       |       | 151     |           |       |       |       |       |  |
| 13579                                  | 2360* | 1930* | 35*     | 2279      | 2.08* | 2.08* | 2.08* | 2.08* |  |

mm (exponents dimensionless) \* fixed - = tied (constant cabin section)  
pchip (monotone cubic) through the knots of each line

**SECTION LAW (each half)**

$|y| / |hw|^n + |(z - zmw) / hz|^m = 1$

hw = half-breadth, zmw = max-breadth WL  
hz = crown - zmw (upper) or zmw - keel (lower);  
n: crown / keel flatness, m: side flatness;  
n = m = 2: ellipse. Section parameter t = polar angle about (0, zmw) in normalised coordinates.

| DEVIATIONS OF THE LOFT FROM THE PILATUS DRAWING (NORMAL DISTANCE, mm) |      |     |      |           |           |                   | REMARK                                                                                           |
|-----------------------------------------------------------------------|------|-----|------|-----------|-----------|-------------------|--------------------------------------------------------------------------------------------------|
| DATASET                                                               | N    | RMS | MAX  | REV A RMS | REV A MAX | WORST AT          |                                                                                                  |
| PROFILE CROWN                                                         | 794  | 1.0 | 12.9 | 109.5     | 141.1     | STA 1060          | max at the cowl front: drawn nose 18 high (drawn prop axis WL 1673; WL 1655 kept)                |
| PROFILE KEEL                                                          | 749  | 2.1 | 11.7 | 178.2     | 333.9     | STA 1060          | visible keel only (not behind wing fairing 5300-7470; strake 10720-11960); max as crown          |
| HALF-BREADTH                                                          | 1164 | 0.9 | 3.1  | 90.8      | 230.7     | STA 12740         | plan view; strake tips STA 11650-11950 ignored                                                   |
| FIN-ROOT LINE                                                         | 161  | 1.6 | 3.4  | 118.1     | 137.5     | STA 12450         | dorsal-fin root line (side view) at the plan fillet half-breadth; STA 9250-12450                 |
| LOWER SILHOUETTE                                                      | 286  | -   | 0.0  | -         | 343.3     |                   | fuselage never below the strake / ventral-fairing silhouette (STA 10700-13560); MAX = protrusion |
| EF1 1200                                                              | 68   | 4.0 | 14.5 | 68.0      | 95.8      | BL 254 WL 1559    | cowl ring only; the chin-inlet crescent is not OML (Stage 3 inlet)                               |
| EF2 2075                                                              | 133  | 0.7 | 1.5  | 72.2      | 134.7     | BL 292 WL 1042    |                                                                                                  |
| FR10 3080                                                             | 132  | 1.9 | 3.8  | 28.7      | 54.2      | BL 135 WL 2162    |                                                                                                  |
| FR12 3580                                                             | 133  | 2.8 | 5.9  | 34.7      | 93.1      | BL 731 WL 2096    |                                                                                                  |
| FR14 4080                                                             | 144  | 3.2 | 6.9  | 56.7      | 130.5     | BL 308 WL 961     |                                                                                                  |
| FR16-30 4590-8230                                                     | 173  | 2.3 | 8.2  | 77.8      | 139.1     | BL 337 WL 965     | constant cabin section, checked at 5 stations 4590-8230                                          |
| FR33 9000                                                             | 225  | 4.3 | 11.4 | 90.2      | 212.3     | BL 618 WL 1444    | keel knee region (flat bottom to tail-cone transition)                                           |
| FR38 9750                                                             | 189  | 1.6 | 3.4  | 116.6     | 295.9     | BL 745 WL 1550    | dorsal fin excluded                                                                              |
| FR38 10800                                                            | 69   | 0.9 | 2.5  | 201.4     | 344.3     | BL 0 WL 2696      | dorsal fin, ventral strakes excluded                                                             |
| FR40 11850                                                            | 116  | 0.5 | 1.2  | 207.3     | 350.4     | BL 403 WL 2263    | dorsal fin, strakes, ventral fairing (BL 121; WL 1606) excluded                                  |
| FR19-FR31 (SHEET 2)                                                   | 840  | 4.3 | 12.1 | 66.0      | 139.8     | FR31; BL 0 WL 939 | phantom cabin section, weight 0.25 (sheet 2 - x +/-15, z +/-45); FR31 lies on the keel knee      |

Targets: profile / half-breadth RMS <= 8, max <= 20; every frame section max <= 20 - ALL MET. REV A = the model before this refit.

| CABIN SECTION NUMBERS           |              |
|---------------------------------|--------------|
| CABIN (STAGE 3)                 | mm           |
| crown / keel WL (STA 4600-8200) | 2769 / 941   |
| OML depth / max breadth         | 1828 / 1688  |
| max-breadth WL                  | 1861         |
| ceiling WL (crown - 40)         | 2729         |
| floor WL (ceiling - 1470)       | 1259         |
| floor WL change vs rev A (1120) | +139         |
| OML width at the floor          | 1545         |
| floor width avail. (- 2 x 85)   | 1175 >= 1300 |
| max inner width (- 2 x 85)      | 1518 - 1520  |

#### NOTES

- Drawn from the loft parameters of model/fuselage.py (knot tables + section law), not from the mesh. Stations = STA mm aft of the datum (3000 ahead of the firewall); WL above the static ground line; BL from the centre line, + starboard.
- Knot values fitted by least squares (drawing/lines, fit.py) to the registered Pilatus NGX model drawing 190.10.40.432: side crown / visible keel, plan half-breadth, frame sections EF1-FR40 (sheet 1) and the sheet-2 phantom cabin section, with smoothness regularisation. Deviation = normal distance, + = ours outside.
- Fixed: spinner tip STA 300, prop axis WL 1655 at the disc (prop clearance 320 kept, the drawing shows 336); thrust line 2 deg nose-down / 2 deg right (drawn), so the spinner tip is at WL 1636 and the spinner base centre 4 mm above the cowl-front ring (R 250 at STA 1044; Stage 3); firewall STA 3000, tail-cone closure STA 13570 at the lower end of the rudder trailing edge; 14,400 overall length is set by the tail bullet.
- Not part of this OML (Stage 3 parts): chin-inlet crescent, dorsal fin, ventral strakes, ventral fairing under the tail cone (FR40, lower silhouette aft of STA 11900), wing-root fairing. Aft of the rudder leading edge (STA 12650 at the keel) the tail cone lies inside the rudder / fin and is trimmed there.
- Changes vs rev A: cabin crown 2630 -> 2769, keel 800 -> 941 (flat from STA 2850), aft body slimmer (half-breadth STA 11000 740 -> 569), tail end 14360 -> 13570, spinner R 290 -> 250, cowl front 950 -> 1044.

## PILATUS PC-12 PRO

### FUSELAGE LINES PLAN

|                                   |     |          |             |            |
|-----------------------------------|-----|----------|-------------|------------|
| DWG NO                            | REV | SCALE    | SHEET       | UNITS      |
| PC12-L1                           | A   | AS SHOWN | A1 · 1 OF 1 | mm         |
| DRAWN                             |     |          |             | DATE       |
| Claude - loftkit parametric model |     |          |             | 2026-09-24 |

RECONSTRUCTION FROM PUBLISHED DATA - NOT FOR CONSTRUCTION